Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/28/2019
PI Name:	Vanapalli, Siva		
Project Title:	Determining Muscle Strength in Space-Flown Caenorhabditis elegans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Lubbock	**State:**	TX
Zip Code:	79409-0001	**Congressional District:**	19
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/19/2015	**End Date:**	04/30/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 4/30/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/12/19) NOTE: Extended to 8/31/2019 per F. Hernandez/ARC (Ed., 3/5/18) NOTE: Extended to 5/18/2018 per F. Hernandez/ARC and NSSC information (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Blawzdziewicz, Jerzy Ph.D. (Texas Tech University)		
Grant/Contract No.:	NNX15AL16G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A major impediment to long-duration space travel is loss of muscle mass and strength during spaceflight. Surprisingly, the small worm Caenorhabditis elegans is a good, cost-effective model for studying the effects of spaceflight upon muscle. Our past experiments on Earth and on the International Space Station (ISS) have shown that the expression of key muscle genes that encode members of a muscle attachment complex is reduced in C. elegans during spaceflight. A key open question is if and how these changes in gene expression result in reduced muscle strength in response to spaceflight. While reduced force production is documented for astronauts, it is yet to be demonstrated for C. elegans. To address this critical knowledge gap, we developed a tiny new device that enables continuous measurement of worm muscle forces. Using this device, on Earth, we have shown that defects in muscle attachment complexes lead to decline in worm muscle strength. Motivated by these past flight data and ground results, we propose to (a) demonstrate the flight-readiness of our miniaturized device for quantitative characterization of muscle strength of C. elegans, (b) test our hypothesis, based on past flight data, that muscle strength decreases in flight as a result of altered muscle gene expression in the worm, and (c) quantify muscle strength loss in spaceflight across multiple worm generations. Successful measurements of muscle strength in flight will make C. elegans a much more complete and comprehensive genomic model for muscle physiology, enabling meaningful translation of discoveries made in C. elegans to astronauts. Our proposed studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies have important spin out implications for sarcopenia (an urgent public health problem on Earth) and may produce a terrestrial model for testing potential therapeutic interventions for muscle strength decline in flight.
Task Progress:	1) We have optimized the conditions for worm culture in the fluorinated ethylene propylene (FEP) bags with culture volumes of 20 mL. We found that an inoculum concentration of 1000 larvae/mL is sufficient for sub-cultures and for animals to grow within the temperature range of 16C – 20C. These culture conditions allow a maximum launch delay of 7 days. 2) We have designed and optimized the NemaFlex chamber design to enable size-based sorting of animals from the unsynchronized cultures present in the FEP bags. We have also developed a protocol for loading worm cultures into this device that now is capable of providing strength data from at least 30 gravid adults. 3) We have conducted multigenerational studies under these culture conditions and measured strength of wild-type and dys-1 animals. Our results show that the strength of wild-type and dys-1 animals do not change over the 7 generations studied. 4) We have tested whether the NemaFlex devices will be free of air pockets during long-term storage. Our data shows that storing NemaFlex devices in Bitran bags with water-soaked gauges keeps the water-loaded NemaFlex devices free of air-pockets. 5) We have visited BioServe in Aug, 2018 and have tested the microscope and camera for imaging. The recorded videos from this flight hardware were found to be suitable for analysis by our software. 6) We have worked with BioServe to build the worm loading apparatus. The basic version of the apparatus has been tested and recommendations have been provided to BioServe for improvements to be made.
Bibliography Type:	Description: (Last Updated: 10/20/2025)
Articles in Peer-reviewed Journals	Rahman M, Hewitt JE, Van-Bussel F, Edwards H, Blawzdziewicz J, Szewczyk NJ, Driscoll M, Vanapalli SA. "NemaFlex: A microfluidics-based technology for standardized measurement of muscular strength of C. elegans." Lab Chip. 2018 Jul 24;18(15):2187-201. https://doi.org/10.1039/c8lc00103k ; PubMed PMID: 29892747; PubMed Central PMCID: PMC6057834 , Jul-2018
Articles in Peer-reviewed Journals	Hewitt JE, Pollard AK, Lesanpezeshki L, Deane CS, Gaffney CJ, Etheridge T, Szewczyk NJ, Vanapalli SA. "Muscle strength deficiency and mitochondrial dysfunction in a muscular dystrophy model of C. elegans and its functional response to drugs." Dis Model Mech. 2018 Dec 4;11(12):dmm036137. https://doi.org/10.1242/dmm.036137 ; PubMed PMID: 30396907; PubMed Central PMCID: PMC6307913 , Dec-2018
Articles in Peer-reviewed Journals	Gaffney CJ, Pollard AK, Deane CS, Cooke M, Balsamo M, Hewitt J, Vanapalli SA, Szewczyk NJ, Etheridge T, Phillips BE. "Worms in space for outreach on Earth: Space life science activities for the classroom." Gravit Space Res. 2018 Dec;6(2):74-82. https://doi.org/10.2478/gsr-2018-0011 (Ed. note--see PDF for actual published article) , Dec-2018